

QUESTION 1

BDM Theory Course: Module – Economics (GA-2)

Read the questions carefully and based on different cost concept associated with Economics and PPF concepts taught in class answer the following questions:

1. Q1 – Q4: 2 marks each
2. Case -1: 6 Marks (2marks/each)
3. Case -2: 6 Marks (2marks/each)

Read questions carefully and then attempt it using Economics Concepts.

Wish you all The Best!

You purchased non-refundable plane tickets for a trip to Mumbai this weekend for ₹5,500/-. You estimate that the trip would give you ₹10,000/- worth of value (enjoyment and benefits). Your friend has just called and informed you that he won 2 free pavilion tickets to the IPL final match between **Gujarat Titans and RCB** (cost ₹8,000/- each) and wants to know if you would like to join him. Calculate the effective value one must gain from the IPL final to justify missing the scheduled flight to Mumbai.

Answer is Correct

Score: 2 / 2

Options

- ☐ A. ₹5500
- ☐ B. ₹16000
- ☒ C. ₹10000
- ☐ D. ₹8000

Correct

 FEEDBACK

₹5500/- is the sunk cost can't be recovered. So effective trade-off cost will be what is the alternative effective value you can lose i.e ₹10,000/-

QUESTION 2

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Read the questions carefully and based on different cost concept associated with Economics and PPF concepts taught in class answer the following questions:

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2. Case -1: 6 Marks (2marks/each)
3. Case -2: 6 Marks (2marks/each)

Read questions carefully and then attempt it using Economics Concepts.

Wish you all The Best!

Technocrat001 , an AI based startup located in Chennai, uses its fixed team of software engineers and graphics processors to develop two products: Autonomous Drone Software and Medical Imaging Analytics. If the startup is **currently operating efficiently on its Production Possibility Frontier (PPF)**, what is the **opportunity cost** of allocating more engineering hours to optimize the Medical Imaging Analytics product?

Answer is Correct

Score: 2 / 2

Options

- ☐ A. The total profit generated by selling both software packages simultaneously.
- ☐ B. The monetary value spent on engineering salaries and server electricity.
- ☒ C. The value or quantity of Autonomous Drone Software updates that are sacrificed.
- ☐ D. Zero, because software code can be duplicated infinitely at no extra cost.

Correct

 FEEDBACK

Because the firm is on the PPF boundary, resources are fully deployed; shifting resources to one product inherently requires reducing the output of the other.

QUESTION 3

BDM Theory Course: Module – Economics (GA-2)

Read the questions carefully and based on different cost concept associated with Economics and PPF concepts taught in class answer the following questions:

1. Q1 – Q4: 2 marks each
2. Case -1: 6 Marks (2marks/each)
3. Case -2: 6 Marks (2marks/each)

Read questions carefully and then attempt it using Economics Concepts.

Wish you all The Best!

A company operating a Software-as-a-Service (SaaS) platform experiences a massive surge in growth driven by increasing **online education, e-commerce, and other digital services along with digital money transaction**. While their **Total Fixed Costs** (such as office rent, core system architecture, and baseline salaries) **remain unchanged**, they are able to distribute these overhead expenses across millions of new subscribers. Which of the following trends is most likely to occur?

Answer is Correct

Score: 2 / 2

Options

- ☒ A. The Average Fixed Cost (AFC) per subscriber will continuously decline toward zero. Correct
- ☐ B. The Marginal Cost (MC) will become exactly equal to the Total Fixed Cost.
- ☐ C. The Average Total Cost (ATC) will begin to increase immediately.
- ☐ D. The Total Variable Cost (TVC) will automatically drop to zero.

 FEEDBACK

Average Fixed Cost is calculated as TFC/Q . As scale (Q) grows larger against a constant numerator, the per-unit fixed cost drops continuously

QUESTION 4

BDM Theory Course: Module – Economics (GA-2)

Read the questions carefully and based on different cost concept associated with Economics and PPF concepts taught in class answer the following questions:

1. Q1 – Q4: 2 marks each
2. Case -1: 6 Marks (2marks/each)
3. Case -2: 6 Marks (2marks/each)

Read questions carefully and then attempt it using Economics Concepts.

Wish you all The Best!

A chip manufacturing company, working on developing hardware for enterprises calculates that at its current output of 10,000 microchips, its Average Total Cost (ATC) is ₹1500.00 per chip. The Marginal Cost (MC) of producing the 10,001st chip is ₹800.00. What will happen to the Average Total Cost if the firm decides to produce that additional chip?

Answer is Correct

Score: 2 / 2

Options

- ☒ A. The ATC will decrease.
- ☐ B. The ATC will increase.
- ☐ C. The Total Cost of production will decrease.
- ☐ D. The ATC will stay exactly at ₹1500.00.

Correct

 FEEDBACK

When the incremental cost of the next unit produced which is marginal cost ---- reduced in this case. Coming down from 1500 to 800. Due to this pulls the current average (ATC) and the overall running average decline.

☐ C. It will drop to zero because a stall can only specialize in one efficient product at a time.

Correct

☐ D. It will decrease because resources must be shifted from tea to coffee production.

 FEEDBACK

According to the Production Possibility Frontier (PPF), a trade-off exists between the two items: reducing the production of Coffee allows for an increase in Tea production, and vice versa. For Ramu to achieve operating efficiency, he must produce at a point directly on the PPF curve.

- ☐ C. The AFC per cup remained exactly the same because rent is a fixed cost.
- ☐ D. The AFC per cup decreased by half because less milk and sugar were wasted.

 FEEDBACK

Average Fixed Cost is calculated as Total Fixed Cost divided by output ($AFC = TFC/Q$). Halving the quantity doubles the resulting value.

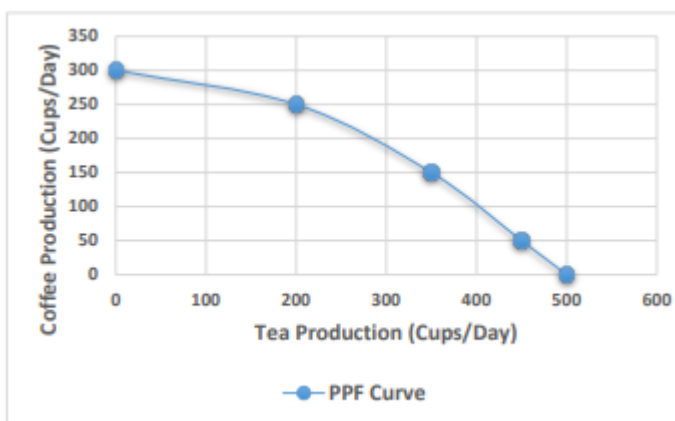
QUESTION 7

The conceptual and numerical problems are based on the hypothetical case. Read the case carefully and based on the Economics concepts taught in class answer the following questions: (Total Marks - 6)

Case Scenario:1**Ramu's Tea Stall at IIT Madras**

Ramu's tea stall was one of the most famous hotspots at IIT Madras, located near the Himalaya Mess on the campus. While he served a variety of snacks, cold drinks, and other refreshments, his tea, coffee, and samosas were the most popular items, consistently generating the highest sales. On any given day, Ramu operates with a fixed set of resources: a specific stall size, two burners, a limited amount of labour hours, and fixed daily supplies of milk, tea leaves, and coffee powder. With these resources, his primary focus is producing two major staple items: Tea and Filter Coffee.

The PPF is a curve that shows the maximum possible output combinations of two goods an economy (or a business like Ramu's) can produce using all available resources efficiently.



Here is how Ramu's daily production frontier looks:

Ramu is currently producing 200 cups of tea per day. His Total Cost (TC) is ₹15,000, of which ₹8000 is his Fixed Cost (FC) for rent and licenses. What is Ramu's Average Variable Cost (AVC) at this level of production?

Answer is Correct

Score: 2 / 2

Options

- ☐ A. ₹30/ cup
- ☐ B. ₹28/ cup

☐ C. ₹40/ cup

☐ D. ₹35/ cup

Correct

 FEEDBACK

Total Variable Cost is $TC - FC = 15000 - 8000 = 7000$. Dividing this by 200 cups gives $7000/200 = 35$.

QUESTION 8

Case Scenario: 2

Tech-based software startup

Data001 is a growing Tech-based software startup with limited resources. Currently, **the team is working at full capacity** and faces a critical resource allocation decision due to its limited resources. The **company must decide how to deploy the team over the next 6 months**. Because team bandwidth is limited, they only have the capacity to fully commit to any one of two available options in the current scenario. Every hour they spend working on an external project is an hour they cannot spend upgrading their own company's systems, and vice versa.

The company must choose one of the following options: **(Total Marks - 6)**

Option A: Prioritize an Urgent Client External Project

I. If they take on this time-sensitive external project, Data001 will earn an immediate, one-time revenue payment of **₹5.0 Lakh**.

II. However, going with option they must delay a critical internal software upgrade. Delaying this upgrade requires paying a vendor extension penalty of **₹1.5 Lakh**.

Option B: Prioritize the Internal Software Upgrade

I. If they complete the internal upgrade now, the new system will immediately reduce the company's operational expenses by **₹10,000 (₹10K)** per month.

II. However, they will not have the bandwidth to take the client contract and will forfeit the **₹5.0 Lakh** revenue.

Suppose Data001's leadership decides to prioritize the urgent client contract (Option A). What is the **opportunity cost** of this decision?

Answer is Correct

Score: 2 / 2

Options

- ☐ A. The engineering time and effort spent fulfilling the client contract
- ☐ B. The ₹1.5 Lakh vendor extension penalty they must pay for delaying the internal upgrade.
- ☒ C. The operational cost savings they would have secured by completing the internal software upgrade.
- ☐ D. The ₹5.0 Lakh one-time revenue they earn from the client

Correct

 FEEDBACK

The operational cost savings they would have secured by completing the internal software upgrade.

QUESTION 9

Case Scenario: 2

Tech-based software startup

Data001 is a growing Tech-based software startup with limited resources. Currently, **the team is working at full capacity** and faces a critical resource allocation decision due to its limited resources. The **company must decide how to deploy the team over the** next 6 months. Because team bandwidth is limited, they only have the capacity to fully commit to any one of two available options in the current scenario. Every hour they spend working on an external project is an hour they cannot spend upgrading their own company's systems, and vice versa.

The company must choose one of the following options: **(Total Marks - 6)**

Option A: Prioritize an Urgent Client External Project

- I. If they take on this time-sensitive external project, Data001 will earn an immediate, one-time revenue payment of ₹5.0 Lakh.
- II. However, going with option they must delay a critical internal software upgrade. Delaying this upgrade requires paying a vendor extension penalty of ₹1.5 Lakh.

Option B: Prioritize the Internal Software Upgrade

- I. If they complete the internal upgrade now, the new system will immediately reduce the company's operational expenses by **₹10,000 (₹10K)** per month.
- II. However, they will not have the bandwidth to take the client contract and will forfeit the **₹5.0 Lakh** revenue.

Suppose Data001 decides to prioritize the urgent client contract to secure the immediate revenue, thereby delaying their internal upgrade for 6 months. What is the **opportunity cost** of this decision over the next 6 months?

Answer is Wrong

Score: 0 / 2

Options

- ☐ A. ₹1.50 Lakh

- ☐ B. ₹10K

- ☐ C. ₹60K

Expected

- ☐ D. ₹2.10 Lakh

Wrong

QUESTION 10

Case Scenario: 2

Tech-based software startup

Data001 is a growing Tech-based software startup with limited resources. Currently, **the team is working at full capacity** and faces a critical resource allocation decision due to its limited resources. The **company must decide how to deploy the team over the next 6 months**. Because team bandwidth is limited, they only have the capacity to fully commit to any one of two available options in the current scenario. Every hour they spend working on an external project is an hour they cannot spend upgrading their own company's systems, and vice versa.

The company must choose one of the following options: **(Total Marks - 6)**

Option A: Prioritize an Urgent Client External Project

I. If they take on this time-sensitive external project, Data001 will earn an immediate, one-time revenue payment of **₹5.0 Lakh**.

II. However, going with option they must delay a critical internal software upgrade. Delaying this upgrade requires paying a vendor extension penalty of **₹1.5 Lakh**.

Option B: Prioritize the Internal Software Upgrade

I. If they complete the internal upgrade now, the new system will immediately reduce the company's operational expenses by **₹10,000 (₹10K)** per month.

II. However, they will not have the bandwidth to take the client contract and will forfeit the **₹5.0 Lakh** revenue.

Suppose Data001 decides to prioritize the urgent client contract to secure the immediate revenue, thereby delaying their internal upgrade for 6 months. What is the **total economic cost** of this decision over the next 6 months?

Answer is Correct

Score: 2 / 2

Options

☐ A. ₹1.50 Lakh

☐ B. ₹5.0 Lakh

☐ C. ₹4.50 Lakh

☒ D. ₹2.10 Lakh

Correct